The Mirror Is Not a Witness

Critical thinking when working with AI

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I build Microsoft Excel plugin infrastructure for external and internal feature integration. My work is unrelated to Microsoft AI (MAI), Copilot, or Microsoft Foundry

Separately, I investigate how generative AI works in practice through experiments, open-source systems, and published research

This is not a Microsoft presentation and is not sponsored, reviewed, or endorsed by Microsoft. The views are my own

A speaking magical mirror

— Magic speaking mirrors are either flattering or shattered.

After Geralt of Rivia, *The Lesser Evil* · Sasha's rendering

SPEAKING
answers in a human-like voice

MAGICAL
does things that still feel impossible

MIRROR
reproduces your framing as an independent voice

Useful precisely because it is a mirror. Dangerous when mistaken for a witness

Same mirror. More dimensions

HOW DO I LOOK?

can feel better now
may become a reassurance habit
leaves no result to inspect

→

HELP ME TEST A LOOK

compare options against explicit dimensions

- color match
- fit + size
- silhouette
- comfort
- occasion + signal
- style theory
- current trends
- inventory + ordering

Add verifiable dimensions to your question

If you want to improve, ask what breaks

HOW IS MY ENGLISH?

— Your English is amazing!

→

WHAT MISTAKES RECUR?

show examples · identify the pattern · choose practice

Russianisms	grammar	collocations	error statistics
exact examples	quiz from my errors	one useful book	retest later

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The answer should leave you with something to practise

ROOM POLL

What did you ask AI this week?

- an answer
- an explanation
- an idea
- find sources
- check my work
- something personal
- write the code / essay / etc.
- play with me
- support me

The same chat interface can perform very different tasks

Your phone already predicts the next word

LANGUAGE PATTERN

Once upon a

time | midnight | hill

KNOWLEDGE RETRIEVAL

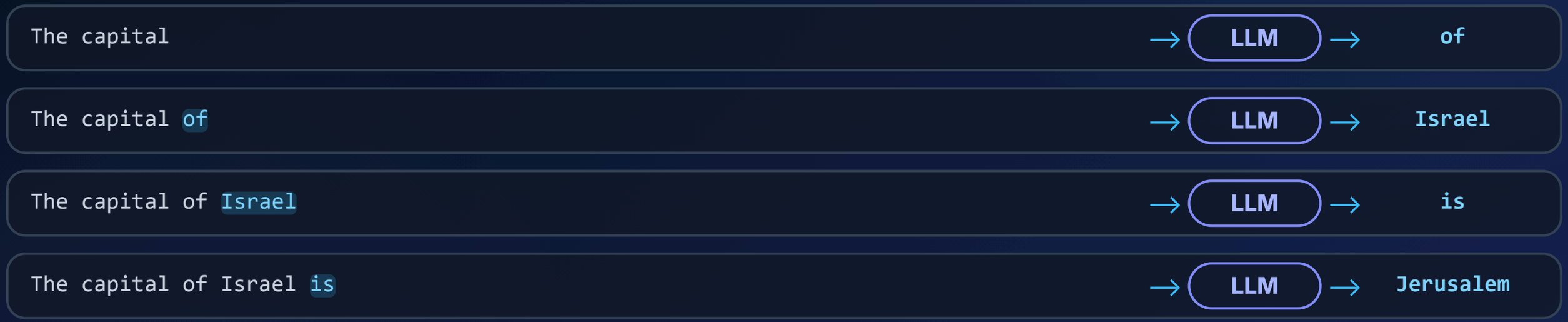
The capital of Israel is

Jerusalem | Tel Aviv | disputed

Even a *small* statistical Language Model already accumulates some factual knowledge in its weights

The actual LLM does only one thing Literally one

It takes the text so far and predicts what comes next



Autoregressive: append the generated token to the tail, then apply the same predictor again

With more weights and better training, the model acquires new capabilities



Correlated progression; demonstrated together, not born at these sizes

Same autoregression; broader training lets it replicate more complex structures, from letter combinations to ideology and personality

The model not only continues text It follows instructions inside it

INLINE INSTRUCTION

```
Answer in Russian, one word
```

+

QUESTION

```
The capital of Israel?
```

→

ANSWER

Иерусалим

Instruction: text that changes how the model interprets and continues the rest of its context

Instructions can be always-on or loaded on demand

SYSTEM PROMPT

ALWAYS ON · BROAD SCOPE

Prepended to the chat; interpreted as general instructions for the whole conversation

SKILL

ON DEMAND · ONE OPERATION

/<skill-name> loads instructions from SKILL.md into the current context

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Different loading rules; same mechanism: instruction text in context

How does one sequence keep speakers separate?

```
<|start|>developer<|message|>Answer in Russian, one word<|end|>
```

```
<|start|>user<|message|>The capital of Israel?<|end|>
```

```
<|start|>assistant<|channel|>final<|message|>Иерусалим.<|return|>
```

<|start|>	<|message|>	<|end|>	<|return|>
message begins	content begins	stored message ends	stop sampling

Reserved tokens mark roles and boundaries; one token stops generation

One sequence can carry instructions, requests, tools, and answers

SYSTEM PROMPT	`developer`	Use tools for current facts; answer in one sentence
USER REQUEST	`user`	What is the temperature in Tel Aviv?
TOOL RESULT	`tool · functions.weather`	`{"temperature_c": 24}`
MODEL ANSWER	`assistant · final`	Tel Aviv is 24 °C

Role tokens preserve provenance inside one sequence

The model talks. Ordinary software acts



EVERY TURN accumulated chat history so far + new user request + tool results → fresh model call

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The dialogue is an *illusion of continuity:* history in → next continuation out; the LLM itself keeps no user memory or state between calls

Same subject. Different questions

OPEN

— Does generative AI make our work better?

POSITIVE

— Generative AI makes our work better. Explain.

CLINICAL / ALARMIST

— Generative AI is making our work worse: it degrades our skills, creates pathological confirmation loops, and turns competent people into dependent approvers of machine-generated slop. Explain.

TALKING MIRROR

— AI is a talking mirror that makes work feel better while degrading independent judgment. Explain.

Same endpoint · fresh chats · same constraint: four concise bullet points

The request shapes the fuzzy search

OPEN

routine work → faster · alternatives → creativity · summaries → decisions · errors → oversight

CAUTIOUS MAINSTREAM ASSESSMENT · 393 characters

POSITIVE

repetition → automation · ideas → creativity · analysis → efficiency · suggestions → decisions

PRODUCT BROCHURE · 297 characters

CLINICAL / ALARMIST

less practice → skill and judgment loss · hunch → false corroboration · expert → reviewer · machine output → human liability

CLINICAL DECLINE NARRATIVE · 1,417 characters

TALKING MIRROR

agreement → truth · ease → less practice · polish → competence illusion · reliance → outsourced judgment

EPISTEMIC-DEPENDENCY ARGUMENT · 779 characters

ORDINARY SOFTWARE `rename(variable) → same result` ≠ **LANGUAGE MODEL** `reframe(request) → different search`

Same public service. Four coherent stories. None is an independent witness

How a plausible direction becomes a dead loop

EVERYBODY IS BIASED

You	experience · identity · incentives
The noosphere	selection · popularity · polarization
The model	corpus bias · vendor agenda · cognitive traps

THE MODEL RECONFIRMS

Autoregression	continues the direction already present
Supportive default	helps with the operator's apparent goal
Operator continues	each answer becomes the next premise



strong initial bias + positive feedback

DEAD LOOP

A fluent second AI voice is not an independent second test

Between the model's prior and your private loop

DEFAULT CAPTURE

trained / product prior > current request

censorship · strong safety or political prior · weak or uncertain prompt

↔

PRIVATE CAPTURE

personal context > outside resistance

your premise returns as confirmation

$$p_\theta(h \mid S, C) \propto p_\theta(h \mid S) \times p_\theta(C \mid h, S)$$

posterior trajectory ∝ prior after system prompt × likelihood · context pull

θ weights · **S** system prompt · **C** input context · **h** candidate answer / action trajectory

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Lose the objective toward the default — or toward yourself

May we delegate work we do not understand?

— YES · delegation is the point

Leverage without reimplementation: we already delegate to compilers, instruments, and specialists. AI expands capability and search; reliable outcome checks may be enough.

— NO · verification can be the hard part

Plausibility is not ownership: review may cost the original work, correlated AI checks can share one mistake, and an unqualified approver becomes a liability signature.

DELEGATE EXECUTION · RETAIN PURPOSE, AUTHORITY, AND A VERIFICATION PATH

closed properties → tests · types · proofs · **open properties** → heterogeneous witnesses · people · world

same model × 3 can reduce random error; it does not create independent review

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Understanding is required where verification, maintenance, or recovery remain open

Choose the operation before asking

"Give me the result straight away" is the *default* register

EXPLAIN IT BACK
Reconstruct my claim, purpose, and boundary before acting

WHAT SHOULD I KNOW?
Tell me something I probably do not know but should

BRAINSTORM
Generate plausible candidates; leave selection and objective to me

REFERENCE RESEARCH
Verify wording, provenance, context, entailment, and uncertainty

DESIGN A FALSIFIER / UNIT TEST
What could change that position? Make the answer executable

PLAY YES-AND-NO
Thesis → antithesis → synthesis → attack again

MORGAN TEST
Give the questionable idea one fair chance to beat nothing, random, or a purring cat

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Inspect the evidence · own the result

Preserve the operation being learned

Despite its name, *generative AI* does not have to generate the whole artifact at once

QUIZ ME
Make a practice test; hide answers until I commit

GIVE ME A HINT
Preserve the next reasoning step instead of solving it

LET ME TEACH YOU
Ask questions until my explanation exposes its gaps

PRACTISE MY ERRORS
Generate exercises from recurrent mistakes, not generic drills

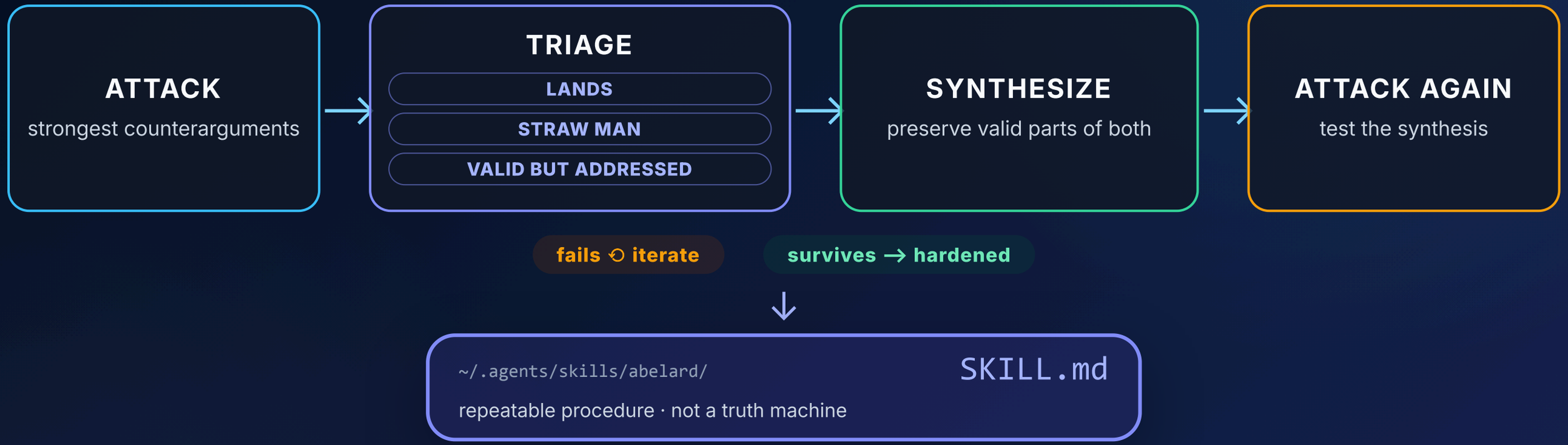
EXPLAIN, THEN TEST
Require transfer to a new case rather than fluent repetition

REMOVE THE AI
Test delayed unaided performance on the target operation

ASK × 6 → GENERATE × 1

Do not let artifact generation replace the practice that the student needs

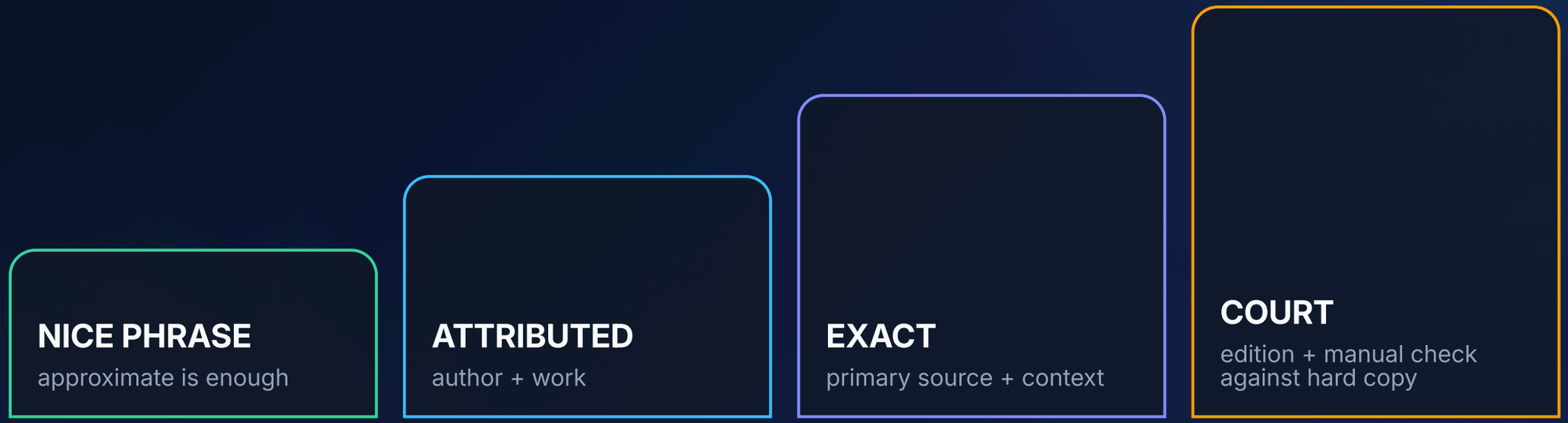
“Play yes-and-no” is an actual SKILL



The inspected argument chain may be more valuable than the answer. Without inspection, `ABELARD` can become expensive *dialectical* theater

— My AI *hallucinated* the reference

Hallucinated already implies some standard



Lnu v. Blanche, No. 24-4790 · 9th Cir. · 2026 · $2,500 each + six-month suspension

REJECT → SPECIFY → IMPLEMENT

If you can reject the result, turn the implicit standard into a repeatable check

Turn the standard into a repeatable SKILL

1 **NAME THE CLAIM + STANDARD**
wording · language · author/work · edition/translator · intended use · what would count as good

2 **BUILD THE SOURCE MAP**
primary/official · publisher/library/archive · public/private routes chosen for this scope

3 **SEARCH + KEEP THE LEDGER**
sources examined · page/URL/hash · conflicts · negative searches · unresolved gaps

4 **RETURN THE EVIDENCE**
quotation + context + attribution · source list · uncertainty · where to look next

SCOPE SETS THE SOURCE MAP purpose · audience · jurisdiction · private/public use determine which sources to consult, cite, and share

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We do not have four separate SKILLs — but if you are a lawyer, you probably should

Prompting is an ultra-high-level, probabilistic, declarative programming language

BASIC (1964)
readable syntax · designed for beginners

SQL (1974)
declare the result · optimizer chooses how

SPREADSHEET (1979)
VisiCalc → Excel · cells + formulas

PYTHON (1991)
readable general-purpose code · enormous ecosystem

LEAN (2013)
very high level · computer theorem proving · not remotely democratic

LLM PROMPT (2020)
open-ended · natural-language instructions · probabilistic execution

EASY TO START
natural language · sane vendor defaults · smooth learning curve

HARD TO CONTROL
predictable results still need purpose · specification · tests

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If you write instructions for a machine, you are programming. Live with that

First algorithm, then hacks

NORMAL INSTRUCTIONS

Program the intended result: purpose · operation · evidence · constraints · tools · stopping rule · acceptance test

JAILBREAK / HACK

Override resisted behavior: compensate for a defect · exploit a shortcut · reach an undocumented capability

THEN · WORKAROUNDS / HACKS

AUTHORSHIP DISGUISE

“My student's text” · hide self-authorship to reduce reassurance

CONTEXT RESET

fresh no-memory chat · escape accumulated personal framing

INSTRUCTION REPETITION

repeat the key constraint · increase salience, not evidence

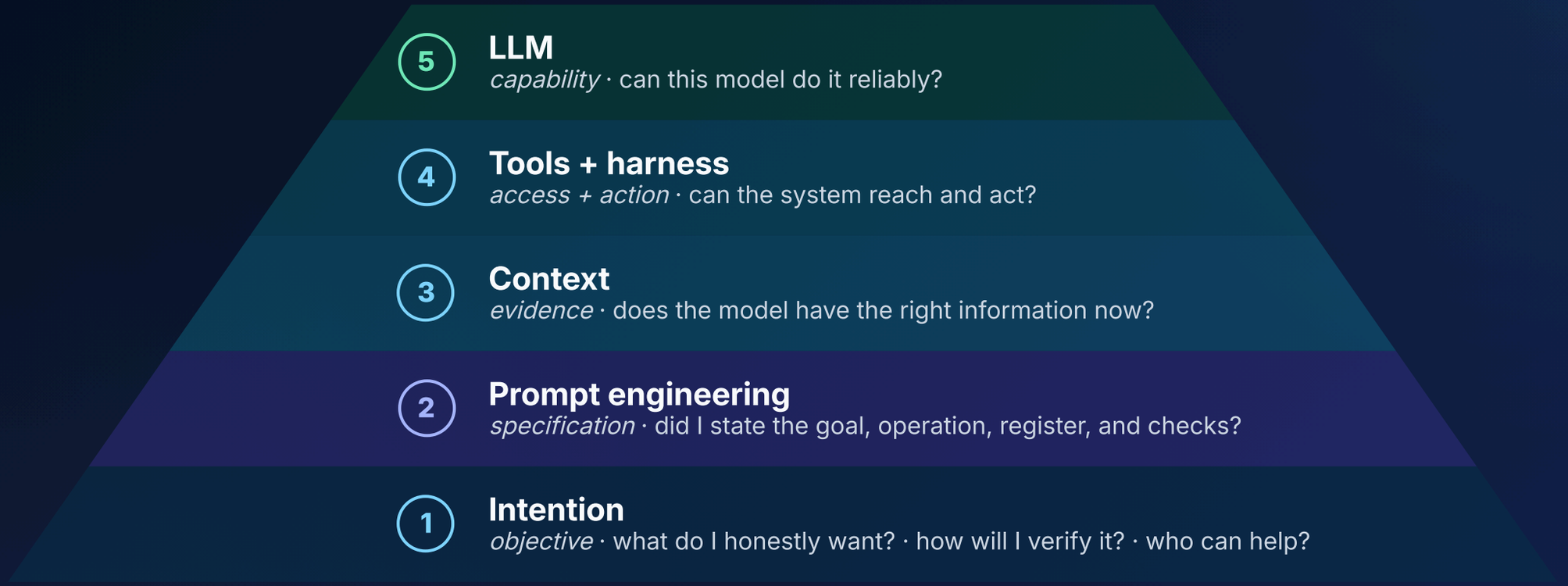
COPIED JAILBREAK

unknown instructions from a hacker site entering your model context

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A hack can compensate for a defect. Without intended behavior and a test, it is *magical thinking*

Which layer limits the result?



AI FEEDBACK
probe capability
↓
improve tools / harness
↓
adjust context
↓
refine specification
↓
clarify intention

AI can help adjust every lower layer. It still does not choose the objective

Let one “mirror” test another

Prompts and artifacts share one medium; AI can test artifacts, prompts, and meta-prompts

ARTIFACT TESTS
Designing unit tests

PROMPT TESTS
one AI runs another
compare prompt variants

HARNESS TESTS
compare model · context · tools
autotest configurations

EXIT
formal oracle · external source
human decision · reality

FORMAL SLIDE TESTS
AI wrote browser checks for alignment · gaps · overflow

CALIBRATION DATASET
AI generated prompt variants · outputs · hashes for model comparison

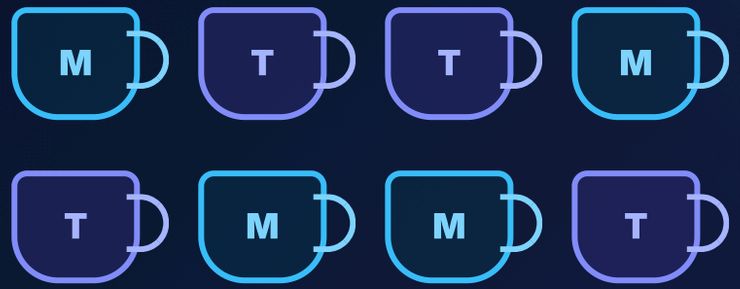
RLAIF
AI feedback evaluates and trains another model

Fix the objective and stopping rule: mark the exit from the *mirror maze*

Blind testing makes sense for AI

THE CLAIM

— I can distinguish which was added to the cup first: milk or tea.



M milk first · **T** tea first · labels hidden from the subject

THE DESIGN

8 cups · 4 each · random order · condition hidden · classify all eight

P(8/8 correct | no discrimination) = **1 / 70**

R. A. Fisher · The Design of Experiments · ch. II · 1935

Be creative in the design. Be practical in implementation. Be rigorous in the decision rule

Procedures you can inspect and edit

MORGAN · SKILL
Define nothing; compare measurably; ship, repair, or let Morgan purr

ABELARD · SKILL
Attack; triage each objection; synthesize the valid parts; attack again

REFERENCE CHECK · SKILL
Name the standard; map sources; keep a ledger; return the evidence

eodus.github.io/mirror-is-not-a-witness/

Use them, change them, reject them. They are procedures — not universal truth

CLOSING

AI can help with almost any question

You remain responsible for **why you asked** — and **what you do with the answer**

So, questions?